



Rigorous analysis of the reflection/transmission at the end-facets of DFB laser diodes

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Abstract. A rigorous approach is presented for evaluation of reflection/transmission at the end-facets of distributed feedback (DFB) lasers. The calculations are based on a rigorous two-dimensional model to find the field distribution inside the periodic region. Moreover, it utilizes an efficient mode-matching procedure which is discussed in the paper. Although the end-facet effects in DFB lasers can rigorously be taken into account by using this approach, the computational efficiency can also be maintained quite high.

Key words: distributed feedback devices, distributed feedback lasers, periodic structures

1. Introduction

Semiconductor lasers are widely used in long-distance communication systems. In particular InGaAsP/InP Multi-quantum-well (MQW) distributed feedback (DFB) laser diodes at 1.5 μm wavelength and especially those with strain compensated QWs are advantageous because of their low threshold current density, improved temperature behavior, narrow line-width, low frequency chirp, and wavelength tunability compared to conventional double-heterostructure diodes with a bulk active layer.

An important feature of a DFB laser is that it can be integrated in the cavity without using cleaved facets (Kogelnik and Shank 1971). Then optical threshold gain in narrow modes does not result from mirror reflectivity as in a resonant cavity (e.g. in Fabry–Perot laser diodes), but rather from an optical wave coupling. This coupling is distributed and causes feedback only near the Bragg wavelength.

Kogelnik and Shank (1969) first introduced the coupled-mode theory and showed that a DFB laser diode without any end-facets, i.e. the reflectivities at both ends can be assumed to be zero, shows two degenerate longitudinal modes outside the stop-band with identical threshold gain values (Kogelnik and Shank 1972). It is known that an asymmetry in the reflectivities and phases of the corrugation grating at the end-facet of a DFB laser can lead to single-mode operation. It has been shown that the lasing characteristics of a

DFB laser not only depend on the geometrical structure of the active region but also on the two end-facet phases. Because of this, much attention has been focused on the end-facet effects in DFB laser diodes (Chinn 1973; Streifer *et al.* 1975; Akiba *et al.* 1983).

To the best of our knowledge, no rigorous and complete theoretical treatment of the reflectivity of the end-facet in DFB laser diodes can be found in the literature. Such a theory should take into account the two-dimensional propagating and non-propagating modes of the structure. Most of the models which have been introduced to calculate the two-dimensional field distribution in DFB structures by now either use crude approximations or consume very large memory and cpu-time resources (Streifer *et al.* 1975; Sato and Hori 1990; Hori and Sato 1990).

The analysis presented here differs from previous ones by using a very general approach to solve the problem. Using a rigorous and efficient model which has been introduced in our previous contributions (Shahabadi and Schünemann 1994; Akbari *et al.* 1999), the two-dimensional field distribution of all propagating and non-propagating modes of the structure in and outside the active region can directly be found. Matching the modes by fulfilling the boundary conditions, the reflection and transmission characteristics of the end-facets can be obtained. The basic ideas of our model are outlined in Section 2 and an efficient mode-matching approach is described in Section 3. In Section 4, we apply the analysis to a few examples of DFB structures and present the field distributions in these structures. Some numerical results of the calculated reflection and transmission coefficients at the end-facets will also be presented.

2. Rigorous and straightforward model for calculating the field distributions in periodic structures

In this section, we review the two-dimensional model for calculating the field distribution in a periodic structure. The complete theory can be found in (Shahabadi and Schünemann 1994; Akbari *et al.* 1999). Fig. 1 shows a general geometry of a periodic structure like that of a DFB laser with four uniform regions and a periodic (grating) layer. The active layer can be considered as a layer with complex dielectric constant. Also illustrated is the two-dimensional model for analyzing the field distribution in the structure. In this model, each homogeneous layer and the periodic layer are modeled by uncoupled and coupled transmission-lines, respectively. The multiport networks T_1 and T_2 are included in the equivalent network in order to model the transformation of the field inside the periodic region to the field outside of it. The transformation factors of the multiport networks T_1 and T_2 are the eigenvectors of a matrix whose elements can easily be calculated from the

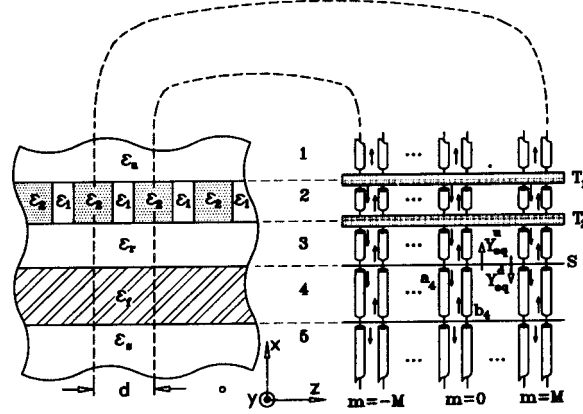


Fig. 1. Geometry of a general DFB structure with rectangular grating and its network model for one period.

Fourier coefficients of the dielectric constant ϵ_r of the grating layer. The advantage of this formulation is that there are not any assumptions involved regarding the shape of the grating. In other words, this formulation is based on a physical treatment of scattering at periodic structures. Maxwell's equations are directly transformed in a way that one yields the telegraphist's equations of a multi-conductor transmission-line. Gratings with continuously varying permittivity and even two-dimensional gratings (binary gratings) can also be treated.

Because of the network formulation, the electric and magnetic field can be replaced by the voltage and current amplitudes of the transmission lines, respectively. To obtain the fields which can be supported by the structure (i.e. in the absence of any wave incident from air or from the substrate), we should find the free-resonant solution of the structure for wave propagation in z -direction. In other words, since the natural frequencies of the network correspond to the propagation constants of the structure, we should determine the zero input response of the network. This is done by dividing the network into two parts, e.g. at plane S in Fig. 1. Admittances looking upward (Y_{eq}^u) and looking downward (Y_{eq}^d) can then be calculated. The continuity of voltages and currents at plane S requires

$$\det(Y_{eq}^d + Y_{eq}^u) = 0, \quad (1)$$

what constitutes a dispersion relation for the guided waves. By solving the dispersion equation, the complex wave numbers of the structure and the amplitudes of the voltages and currents of the transmission-lines in each layer (in other words: the two-dimensional field distribution) can be found.

3. Efficient procedure to analyze end-facet effects

At the end-facet of a DFB laser, there is discontinuity for the waves guided by the periodic region. The end section can consist of a slab waveguide, AR materials, air and/or a combination of them. The guided modes of the structure travel into z -direction so that they are normally incident on the discontinuity. It is known that, when a guided or surface wave is incident normally on a discontinuity (for example a step junction), the boundary condition problem is two-dimensional and an incident TE-wave generates reflected and transmitted waves of TE polarization only. For oblique incidence, on the other hand, the boundary value problem becomes three-dimensional and TE–TM coupling is produced at the discontinuity (Ogusu *et al.* 1979; Peng and Oliner 1981). This means that an incident wave of a given polarization now also produces reflected and transmitted waves of the opposite polarization. This situation is not of interest in our analysis, so that we need only consider TE waves for fulfilling the boundary condition.

As a first step, we consider a periodic structure connected to an infinitely long slab waveguide (see Fig. 2). The active layers are taken into account as the layers with complex dielectric constant. The field distribution in the slab waveguide (E_y^s, H_x^s) can be found using a simplification of the method presented in the previous section. In the connecting waveguides, there is no periodic layer and consequently no coupled transmission-lines are to be included in the model. These structures can be modeled by considering transmission-lines cascaded in x -direction. The boundary condition at the discontinuity requires that the total tangential field components are continuous. To fulfill it, we need all discrete and continuous modes of both structures (note that an open structure also shows continuous modes (Marcuse 1970)).

Finding an analytical form for the field distribution in the periodic structure is not a simple task and there is no rigorous method available as of now.

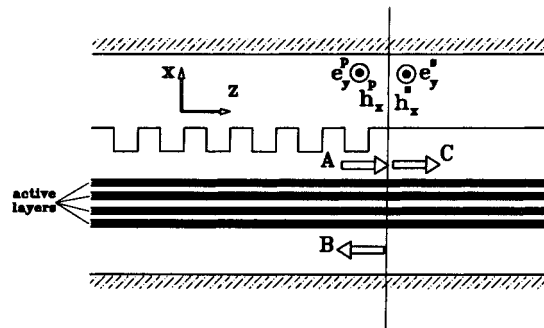


Fig. 2. Boundary condition at the interface between a periodic structure and a slab waveguide.

It is customary, however, in this class of problems (Solbach and Wolff 1978; Mittra *et al.* 1980) to discretize the continuous modes by placing perfectly conducting plates above and below the structure. These plates are placed far above and far below the guiding dielectric region so that they only negligibly influence the properties of modes guided by the structure. In DFB lasers, one also finds these conducting plates in order to inject the current. In addition to the first guided mode, the structure with the conducting plates supports an infinite number of discrete higher-order modes, some of which are below cutoff (i.e. non-propagating). We must also consider these non-propagating modes to completely describe the boundary condition. These modes store energy in the vicinity of the discontinuity.

The perfectly conducting plates can be modeled by short circuits placed at the highest and lowest transmission-lines in Fig. 1. The electric field $E_y(x)$ for all modes can be found using the network model for both the periodic and the slab regions. The tangential magnetic field $H_x(x)$ can be obtained from

$$\nabla \times E(x, z) = -j\omega\mu_0 H(x, z) \implies H_x(x, z) = \frac{-j}{\omega\mu_0} \frac{\partial E_y(x, z)}{\partial z}. \quad (2)$$

We define vectors e_y and h_x as vectors whose elements are normalized and orthogonal components of E_{yi} and H_{xi} , the first- and higher-order modes of the structure:

$$e_y = \begin{bmatrix} E_{y1} \\ E_{y2} \\ \vdots \\ E_{yn} \end{bmatrix}, \quad h_x = \begin{bmatrix} H_{x1} \\ H_{x2} \\ \vdots \\ H_{xn} \end{bmatrix}. \quad (3)$$

In Fig. 2, the A , B and C are vectors of dimension n whose elements constitute the amplitudes of the modes for the incident, reflected and transmitted wave at the discontinuity, respectively. At the interface, we should have

$$(A + B)e_y^p = Ce_y^s, \quad (4)$$

$$(A - B)h_x^p = Ch_x^s. \quad (5)$$

From Equations (4) and (5) and drawing attention to the fact that the elements of the vectors e_y and h_x are orthogonal, we can write

$$C = (A + B)M^e = (A - B)M^h, \quad (6)$$

where M^e and M^h are the transforming matrices which are found to read as

$$M^e = \begin{bmatrix} \frac{\int_{-\infty}^{+\infty} E_{y1}^p E_{y1}^{s*} dx}{\int_{-\infty}^{+\infty} E_{y1}^s E_{y1}^{s*} dx} & \dots & \dots & \frac{\int_{-\infty}^{+\infty} E_{y1}^p E_{yn}^{s*} dx}{\int_{-\infty}^{+\infty} E_{yn}^s E_{yn}^{s*} dx} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\int_{-\infty}^{+\infty} E_{yn}^p E_{y1}^{s*} dx}{\int_{-\infty}^{+\infty} E_{y1}^s E_{y1}^{s*} dx} & \dots & \dots & \frac{\int_{-\infty}^{+\infty} E_{yn}^p E_{yn}^{s*} dx}{\int_{-\infty}^{+\infty} E_{yn}^s E_{yn}^{s*} dx} \end{bmatrix}, \quad (7)$$

and

$$M^h = \begin{bmatrix} \frac{\int_{-\infty}^{+\infty} H_{x1}^p H_{x1}^{s*} dx}{\int_{-\infty}^{+\infty} H_{x1}^s H_{x1}^{s*} dx} & \dots & \dots & \frac{\int_{-\infty}^{+\infty} H_{x1}^p H_{xn}^{s*} dx}{\int_{-\infty}^{+\infty} H_{xn}^s H_{xn}^{s*} dx} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\int_{-\infty}^{+\infty} H_{xn}^p H_{x1}^{s*} dx}{\int_{-\infty}^{+\infty} H_{x1}^s H_{x1}^{s*} dx} & \dots & \dots & \frac{\int_{-\infty}^{+\infty} H_{xn}^p H_{xn}^{s*} dx}{\int_{-\infty}^{+\infty} H_{xn}^s H_{xn}^{s*} dx} \end{bmatrix}. \quad (8)$$

Using Equation (6), we have

$$A(M^h - M^e) = B(M^e + M^h), \quad (9)$$

and consequently

$$B = A[(M^h - M^e)(M^h + M^e)^{-1}]. \quad (10)$$

By specifying the incident field A , we obtain the total reflected field B , and from $C = (A + B)M^e$ the total transmitted field at the discontinuity can also be obtained. It must be mentioned that the size of the matrices M^e and M^h is determined by the number of the modes which we take into account for mode-matching. In most cases, this number is less than 20 so that the inversion of matrix $(M^e + M^h)$ in Equation (10) can be done fast.

For the case that the slab waveguide shows finite length, the problem can be solved in the same way. In this case, we assume as DFB laser a slab

waveguide with length l_s connected to a periodic structure. The other end of the slab is open so that the wave is radiated into air (see Fig. 3). Then one must deal with radiation from an open-ended wave-guide. Open-ended waveguides have become increasingly important in the last twenty years, particularly in connection with integrated optics and millimeter-wave integrated circuits. The widely used approach to this problem is to expand the scattered field in terms of discrete surface-wave modes and of continuously radiating modes, and then to match the tangential field components at the interface of the discontinuity (Mahmoud and Beal 1975; Brooke and Kharadly 1976).

Another rigorous approach is to utilize the Wiener–Hopf technique, which is considered to be one of the most powerful analytical methods for treating boundary value problems which deal with semi-infinite boundaries (Mittra and Lee 1976). Since in this method, the region is separated into two parts (inside and outside of the dielectric guides), it would be suitable for treating open-type problems. The accuracy of the final results, however, depends on the branch cut integrals related to the radiated field, as well as on the truncation of infinite-dimensional algebraic equations associated with the discontinuities of the media at the junction of the discontinuity.

The method based on the Wiener–Hopf technique and the other ones need a very complicated mathematical calculation and hence very large cpu-time. They are not suitable to establish an efficient procedure in the case of DFB lasers. In a DFB laser, the field in the structure is confined to the active layers and decreases exponentially towards the cladding layers. At a distance far from the active layer, the magnitude of the field is almost zero, with very good approximation. Then we can consider the modified structure of Fig. 4 to find the radiation pattern from the end-facet. In this structure, we have

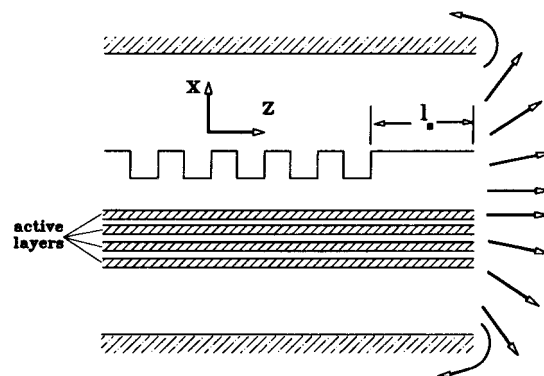


Fig. 3. A slab waveguide with length l_s connected to a periodic structure, the other end of the slab couples to air.

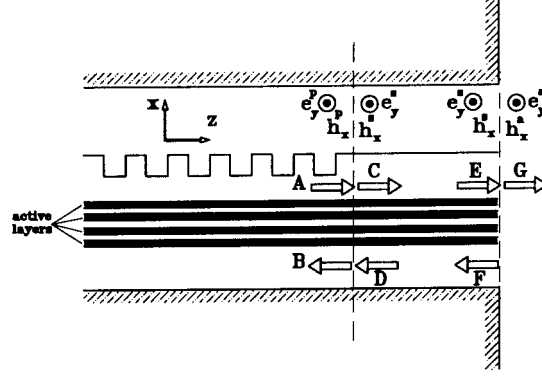


Fig. 4. A proposed structure to calculate the end-facet effects of a DFB structure.

introduced two vertical conducting plates at the end of the slab waveguide. This is equivalent to neglecting the radiation outside the metal sheets in negative z -direction. The radiated field in air has a continuous spectrum and can be represented by

$$e_y(x, z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} E(\beta_x) e^{-\gamma z} e^{-j\beta_x x} d\beta_x, \quad (11)$$

where $\gamma = (\beta_x - k_0)^{1/2}$ and the integration contour is along the real axis from $-\infty$ to ∞ . At the interface of the slab waveguide and air, the radiated field (11) can be written as

$$e_y(x) = \lim_{M, N \rightarrow \infty} \frac{1}{2\pi} \sum_{n=-M}^N E(\beta_{xn}) e^{-j\beta_{xn} x} \Delta\beta_{xn}, \quad \Delta\beta_{xi} = \beta_{xi} - \beta_{x(i-1)}. \quad (12)$$

In this equation the propagation constant β_x is a continuous variable but for presented numerical calculations we approximate it as a set with finite number of elements. $([k_{x\min}, \dots, k_{x\max}])$. The amount of maximum (or minimum) and number of the elements of this set determine the accuracy of the output results and must be found for the determined structure. We define orthonormal vectors e^a and h^a as

$$e^a = \frac{1}{2\pi} \begin{bmatrix} e^{-jk_{x\min} x} \Delta k_x \\ \vdots \\ \vdots \\ e^{-jk_{x\max} x} \Delta k_x \end{bmatrix}, \quad (13)$$

$$h^a = \frac{1}{2\pi} \begin{bmatrix} y_{x_{\min}} e^{-jk_{x_{\min}} x} \Delta k_x \\ \vdots \\ y_{x_{\max}} e^{-jk_{x_{\max}} x} \Delta k_x \end{bmatrix}, \quad (14)$$

where $\Delta k_x = k_{xi} - k_{x(i-1)}$ and y_i is the admittance of the i th plane wave. The electric and magnetic field in air are linear functions of the elements of e^a and h^a , respectively.

We assume that the electric and magnetic fields as well as the propagation constants in both the periodic part (i.e. e^p, h^p , and γ^p , p means periodic) and in the slab waveguide (i.e. e^s, h^s , and γ^s , here s means slab) have already been calculated by the model discussed in Section 2. The boundary conditions then lead to (see Fig. 4)

$$(E + F)e_y^s = Ge_y^a, \quad (15)$$

$$(E - F)h_x^s = Gh_x^a, \quad (16)$$

$$(A + B)e_y^p = (C + D)e_y^s, \quad (17)$$

$$(A - B)h_x^p = (C - D)h_x^s. \quad (18)$$

In the slab waveguide, we have $E = CS$ and $D = FS$. The vectors A, B, C, D, E, F and G are vectors of dimension n for the amplitudes of the waves and S is a diagonal matrix with elements $[e^{-j\gamma_i^s l_s}]$, l_s refers to the length of the slab region. If the amplitudes of the incident waves A are known, the other amplitudes (B, C, D, E, F , and G) can be found from the last six equations using some mathematical manipulation (see Appendix I). When knowing the coefficients B , we can determine the coefficients for all reflected modes at the interface between the slab waveguide and the periodic region. The amplitudes C, D, E and F describe the total field inside the slab waveguide. Using vector G , we obtain the waves transmitted or radiated into air.

This mode-matching method which uses the presented rigorous network model to find the 2-D field distribution in the periodic structure, can be applied to study the effect of the end-facet effects in DFB lasers. In the next section, some simulated results will be given which have been obtained from a computer program using Matlab language. It must be mentioned that the computation time for all cases was very short.¹

¹A PC with Pentium 300 has been used to compute the results. For example, the cpu-time to find the first 20 transversal modes of the active region of the chosen MQW-DFB laser consisting of 27 layers was less than 3 min, if the propagation constants are calculated with a precision of 10^{-3} .

4. Numerical examples

The end-facet effects in DFB lasers are illustrated numerically by applying our model for periodic structures combined with the accurate mode-matching procedure presented above. As a first example, we will present the field distribution in an MQW-DFB laser using the model illustrated in Fig. 1. The chosen MQW laser structure shows 10 active quantum well layers (layers 4, 6, ..., 22 with the thickness 7.8 nm and dielectric constant 3.595) each of which is buried between two barrier layers (passive quantum well layers 3, 5, ..., 23 with thickness 4.8 nm and dielectric constant 3.3381). The lasing wavelength amounts to $\lambda_o = 1532$ nm. The grating has a period of 235 nm and a duty cycle of 35%. The rectangular grating is in layer no. 26 with thickness of 28 nm and its dielectric constant varies periodically in z -direction between 3.165 and 3.256 (35% of a period has been filled with $\epsilon_r = 3.165$). Table 1 summarizes the other parameters of this structure.

This Structure has 10 propagating modes. The normalized distributions of the transversal electric field of the first, second and third mode are presented in Fig. 5A. The propagation constants for these three modes are $3.238836k_o$, $3.090666k_o$ and $3.047041k_o$, respectively, ($k_o = 2\pi/\lambda_o$ where $\lambda_o = 1532$ nm). The 2-D field distribution for this structure assuming an infinite length has been illustrated in Fig. 5B. (This Fig. shows the field distribution inside the structure only for 4 periods.) From the total 2-D distribution of the electric field, we can calculate the confinement factor. This factor, which represents the fraction of the total intensity in the active region of the laser, is defined according to

$$\Gamma = \frac{\int_A \int_{\text{active}} E_y^2(x, z) dx dz}{\int_A \int_{-\infty}^{\infty} E_y^2(x, z) dx dz}, \quad (19)$$

with A period of the grating. The confinement factor and the coupling coefficient κ are two parameters which are very important for a dynamic analysis of MQW lasers. Up to now, some methods to determine the coupling coefficient of the periodic structure have been introduced. To our best knowledge, however, these calculations were based on either a one-dimensional model or on an approximate method to calculate this parameter

Table 1. Parameters of the simulated MQW-DFB structure

Layer	1	2	24	25	26	27
Thickness (nm)	1000	40	57	45	28	1000
Dielectric constant	3.162	3.381	3.381	3.165	3.256	3.165

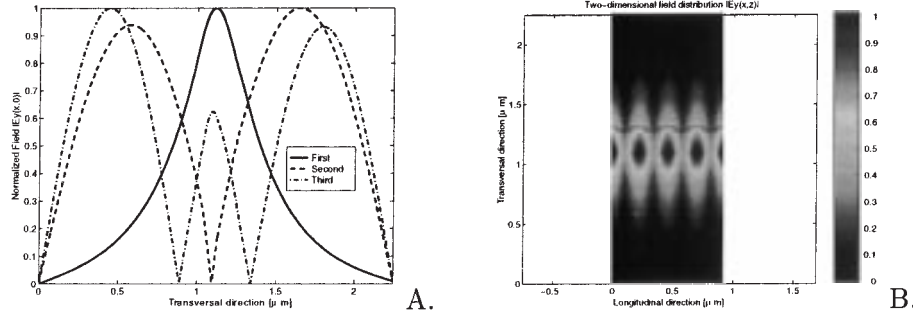


Fig. 5. (A) Normalized transversal electric field ($|E_y(x,0)|$) for the first, second and third propagating mode of the chosen MQW-DFB. The propagation constants of these modes have been given in text. (B) 2-D electric field distribution inside the investigated MQW structure for first order of the grating. The visible standing wave shows the existence of backward waves which are coupled with the forward waves in the structure.

(Streifer *et al.* 1975; Hardy 1984). The rigorous 2-D method presented here can determine the coupling coefficient of the periodic structure for any grating shape. The thickness of the grating, which plays a limiting role in the other methods, has no influence in our modeling. The effective index n_{eff} , the confinement factor Γ and the coupling coefficient κ of the structure amount to $n_{\text{eff}} = 3.25276$, $\Gamma = 0.185774$ and $\kappa = 68 \text{ cm}^{-1}$, respectively.

Now we study the end-facet effect of an MQW laser with parameters summarized in Table 1. There is a slab waveguide at the end of the MQW laser with the same material as in the active region. The field radiates into air from the end of the slab waveguide. (The geometry of the structure is illustrated in Fig. 4.) It must be mentioned that the slab waveguide can consist of arbitrary material combinations. For the analysis we only need to know the transversal field distribution of the electric and magnetic fields of the structures.

The first mode of the MQW laser impinges on the discontinuity between the periodic and the slab waveguide. At the discontinuity, all modes of the structure will be excited and contribute to the reflected mode. For these calculations we have considered 25 modes of grating and slab structures (10 of these total 25 modes are propagating and the others are non-propagating modes). In the air region we truncated the unbounded and continuous variable β_x at $[-10k_0, 10k_0]$. This interval is discretized with step $0.1k_0$, it means 200 surface waves have been considered to model the radiated field in air. Fig. 6A shows the incident and the total reflected waves at the discontinuity. That part of the reflected wave which is related to the first mode of the structure has also been illustrated in the figure. The wave radiated into air at the interface slab-air is depicted in Fig. 6B.

Fig. 7A shows a 2-D visualization of the field at the same conditions described above. The effect of the length of the slab waveguide on the total

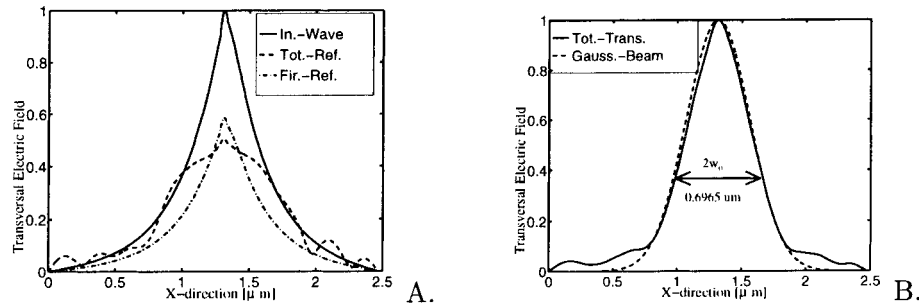


Fig. 6. (A) Incident wave, total reflected wave taking all modes into account, and reflected partial wave just taking the first mode into account. (B) Radiated pattern in air for all modes. For the grating and slab structures 25 modes (10 of them are propagating modes) and for the air region 200 surface waves have been considered.

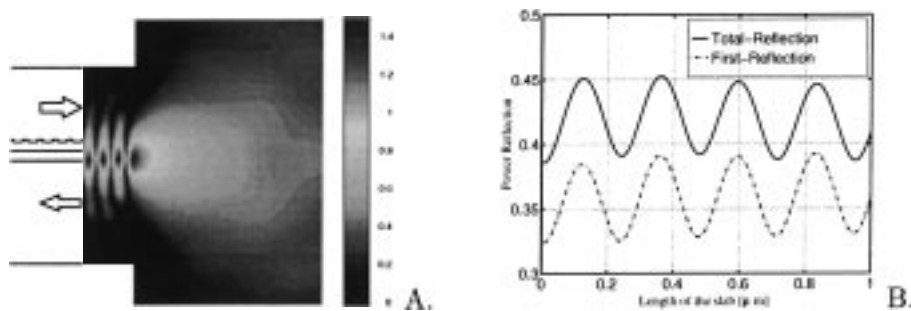


Fig. 7. (A) 2-D field distribution. (B) Total power reflection coefficient and that part of it which is related to the first mode of the structure versus length of the slab waveguide. For the grating and slab structures 25 modes (10 of them are propagating modes) and for the air region 200 surface waves have been considered.

power reflection and that part of it which is related to the first mode of the structure are presented in Fig. 7B. By knowing the vectors of the reflection and transmission characteristics of the structure, the effect of the end-facet phases of MQW-DFB lasers can rigorously be considered.

5. Conclusions

In this paper, we have introduced a rigorous model to calculate the 2-D field distribution in the periodic structure of MQW-DFB lasers. This model is very efficient and is able to calculate the field characteristics of MQW structures (e.g. consisting of 30 layers) in very short cpu-time. It is further capable of determining the field characteristics for arbitrarily formed grating grooves (e.g. for triangular or sinusoidal shape). The field distribution in periodic

structures consisting of a grating with two different periods is presently under investigation.

Using a complete mode-matching procedure, the end-facet effects of MQW-DFB lasers with any arbitrarily formed facet shape can be efficiently studied with the model. Based on a correct and complete definition of the end-facet phases, the static and dynamic characteristics of MQW lasers can almost rigorously be determined. The necessary computer requirements on storage and cpu-time are low.

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Appendix I

Calculation of the vectors B , C , D , E , F , and G

We define the matrices $M_{P-S}^e, M_{P-S}^h, M_{S-A}^e$ and M_{S-A}^h like the matrices M_e and M_h in Equations (7) and (8). The indices $P-S$ and $S-A$ are used to mark the interfaces periodic-slab and slab-air, respectively. Rewriting Equations (14)–(17),

$$(C + D) = (A + B)M_{P-S}^e, \quad (20)$$

$$(C - D) = (A - B)M_{P-S}^h, \quad (21)$$

$$G = (E + F)M_{S-A}^e, \quad (22)$$

$$G = (E - F)M_{S-A}^h, \quad (23)$$

$$E = CS, \quad (24)$$

$$D = FS. \quad (25)$$

From Equations (20) and (21), we have

$$C = 0.5[A(M_{P-S}^e + M_{P-S}^h) + B(M_{P-S}^e - M_{P-S}^h)], \quad (26)$$

$$D = 0.5[A(M_{P-S}^e - M_{P-S}^h) + B(M_{P-S}^e + M_{P-S}^h)]. \quad (27)$$

From Equations (22)–(25), it follows

$$F = CS(M_{S-A}^e - M_{S-A}^h)(M_{S-A}^e + M_{S-A}^h)^{-1}, \quad (28)$$

$$D = CS(M_{S-A}^e - M_{S-A}^h)(M_{S-A}^e + M_{S-A}^h)^{-1}S. \quad (29)$$

From Equations (29), (26) and (27), we obtain

$$\begin{aligned} B = & A[(M_{P-S}^e + M_{P-S}^h)S(M_{S-A}^e - M_{S-A}^h) \\ & \times (M_{S-A}^e + M_{S-A}^h)^{-1}S - (M_{P-S}^e - M_{P-S}^h)] \\ & \times [(M_{P-S}^e + M_{P-S}^h) - (M_{P-S}^e - M_{P-S}^h)S \\ & \times (M_{S-A}^e - M_{S-A}^h)(M_{S-A}^e + M_{S-A}^h)^{-1}S]^{-1} \end{aligned} \quad (30)$$

By having the vector B from Equation (30), the vectors C and D can be obtained from Equations (26) and (27). The vectors G , E and F will then be obtained from Equations (23), (24) and (25).

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